

## Topic quiz: Power and efficiency

IBDP Physics SL · Grade 11 · A.3 Power and efficiency · 10 multiple-choice questions

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Name:

Class:

Date:

**How to use.** Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use  $g = 9.81 \text{ m s}^{-2}$  and show any working next to the question.

**Check yourself online:** the same quiz with instant feedback and explanations — <https://physics.taneski.com/quiz-power-efficiency/>

[illegible]

### Question 1

A 60.0 kg student runs up stairs with a vertical height of 3.0 m in 4.0 s. What is the useful power output?

- (A) 45.0 W                      (B) 441 W  
(C) 1770 W                    (D) 7060 W

## Question 2

A car travels at a constant  $25 \text{ m s}^{-1}$  against a total resistive force of  $800 \text{ N}$ . What power does the engine deliver to the wheels?

- (A) 0 W, because the resultant force is zero
- (B) 32 W
- (C) 800 W
- (D) 20 kW

### Question 3

A motor raises an 800 kg lift at a constant speed of  $1.5 \text{ m s}^{-1}$ . What is the useful power output of the motor?

- Ⓐ 0.90 kW                      Ⓑ 1.2 kW  
Ⓒ 5.2 kW                        Ⓓ 11.8 kW

### Question 4

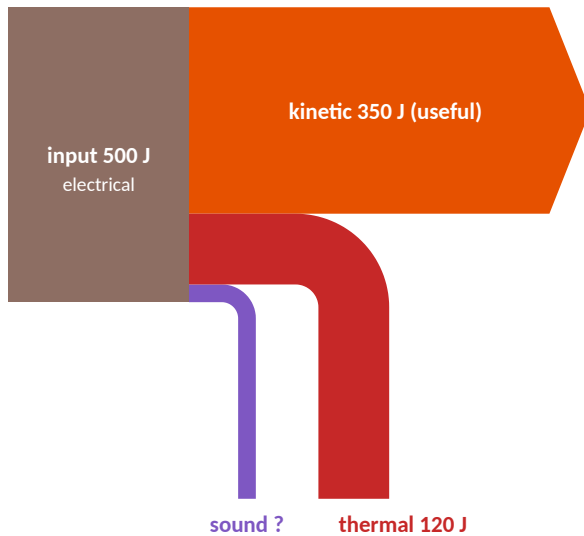
An electric motor uses 3.5 kJ of energy to lift a 150 N load through 7.2 m. What is its efficiency?

- (A) 3.1 %                      (B) 31 %  
(C) 69 %                      (D) 324 %

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### Question 5

The Sankey diagram shows the energy transfers in an electric motor every second. What is the missing sound energy and the efficiency of the motor?



- |                                  |                                 |
|----------------------------------|---------------------------------|
| (A) Sound 30 J, efficiency 70 %  | (B) Sound 30 J, efficiency 30 % |
| (C) Sound 150 J, efficiency 70 % | (D) Sound 30 J, efficiency 74 % |
- 

### Question 6

A ball is dropped from 1.20 m and bounces back up to 0.84 m. What percentage of its energy does it keep after the bounce?

- |          |           |
|----------|-----------|
| (A) 30 % | (B) 70 %  |
| (C) 84 % | (D) 143 % |
- 

### Question 7

Petrol has an energy density of  $46 \text{ MJ kg}^{-1}$ . A car carries 40 kg of petrol, its engine is 25 % efficient, and at  $25 \text{ m s}^{-1}$  it needs 20 kW of useful power. About how far can it travel on this petrol?

- |            |             |
|------------|-------------|
| (A) 23 km  | (B) 144 km  |
| (C) 575 km | (D) 2300 km |
- 

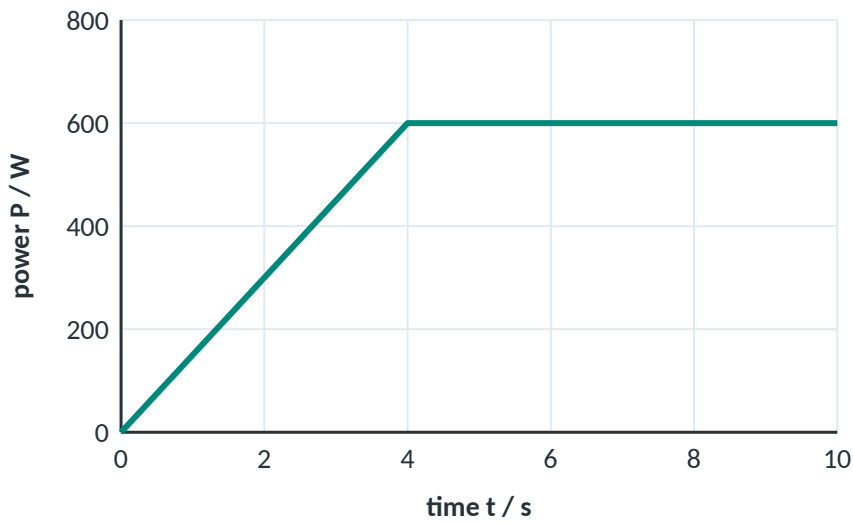
### Question 8

Why can no machine have an efficiency greater than 100 %?

- |                                                                                                |
|------------------------------------------------------------------------------------------------|
| (A) The useful energy out cannot be more than the total energy in, because energy is conserved |
| (B) Machines always need some input power                                                      |
| (C) Friction always wastes exactly half the energy                                             |
| (D) Power is always lost faster than energy                                                    |

### Question 9

The graph shows the power output of a motor. How much energy does it transfer in the 10 s?



- (A) 60 J                      (B) 3600 J  
(C) 4800 J                  (D) 6000 J

### Question 10

A 1200 kg car drives up a  $5.0^\circ$  slope at a constant  $15 \text{ m s}^{-1}$  against a resistive force of 600 N. What is the power output of the engine?

- Ⓐ 9.0 kW                      Ⓑ 15.4 kW  
Ⓒ 24.4 kW                    Ⓓ 185 kW

End of quiz • Score: \_\_\_\_\_ / 10